United States Federal Spending in Georgia

Seth Brice
April 2, 2026

Data Mining M3

I. Dataset Description

The data set used in this study comes from "USASPENDING.gov", a website that collects and holds records of awards given to each state since 2008 [1]. It breaks down awards into multiple categories, including contracts, contract IDVs, grants, loans, direct payments, and other award types (Figure 1). Each award has several pieces of data giving context to the award, including the recipient name, awarding agency, amount granted, etc.

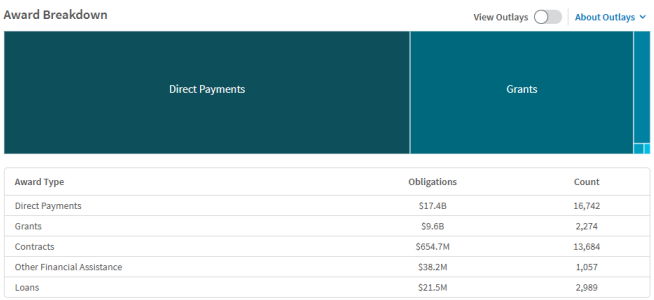

Fig. 1. A graph from "USASPENDING.gov" detailing the sum of all transaction types made to the state of Georgia from fiscal year 2008 to 2026.

This study specifically uses the web page that contains records of all federal awards granted to Georgia. This study only takes into account awards in 2025 totaling over 20,000,000 dollars, as this is the most relevant area for considering current spending. The primary attributes of each data point that will be examined are the recipient name, the awarding agency, the award type, the award subtype, the award description and the obligated amount (Table 1). The obligated value of the award was chosen over the outlay amount due to a significant amount of missing outlays present in the dataset.

While the obligated amount may reduce clarity on where federal money is going, it will still provide context into the interests of the federal government and their spending habits. There may also be a quality issue in the description of each award, as some are more clear than others, while some barely provide a description at all. For this reason, the description is not as considered for analysis as the rest of the data, however it is still provided to give context to each award. After sectioning out the most important portion of time, and carefully considering which pieces of data are most relevant to the study, the data set used contains 1,478 rows of entries, each with 6 columns of data.

transactions	
transaction_id	string
recipient	string
awarding_agency	string
award_type	string
award_subtype	string
description	string
obligations	float

TABLE I
Data schema describing the breakdown of each transaction and its relevant data. This study primarily deals in string values, with the exception of obligations, which are recorded as a float value.

II. Discovery Questions

This data mining project aims to answer the following questions about U.S. federal spending:

- What combinations of awarding agencies, recipients, and award types frequently occur in different sized regions within Georgia?
- "What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?"
- What characteristics distinguish anomalous high-value transactions from normal federal spending in Georgia?

In a time where U.S. citizens desire to know more than ever where their tax dollars are being used, answering these questions is an important task. It is vital to provide clarity and rationale for the average tax payer that their money is going to secure areas that benefit and strengthen their community. Unfortunately in the past few years, waves of questionable and fraudulent programs benefiting off of government subsidy have been brought to light, causing restlessness and conflict in America. Examining the state of Georgia should either provide reassurance to its citizens that their taxes are still primarily going to good places, or will expose an underlying problem within the state in need of correction.

III. Planned Techniques

To begin the process, each transaction made within the relevant time period will be recorded within a data matrix

created using Python code (Figure 2). Once the data has been

```
transactionMatrix = [
    ["FA862506C6456","Lockheed Martin Corp","Department of Defense","Contract","Definitive Contract", "7,261,932,653.67"],
    ["2505GASMAP","GA Department of Community Health","Department of Health and Human Services","Grant","Block Grant", "12,849,962,570.00"],
    ["SLFRP1029","State of GA","Department of the Treasury","Direct Payment","Direct Payment for Specified Use", "4,853,511,033.91"],
    .
    .
    .
    .
    .
    .
    ]
    ]
```

Fig. 2. Sample data matrix demonstrating what the matrix will look like when it is recorded within the python script. This example demonstrates a recorded contract, a grant, and a direct payment.

consolidated into a table, two primary data-mining techniques will be used to analyze it. First, to discover what combinations of transaction characteristics occur in different regions in Georgia, the association rule will be implemented using Apriori. The data will be downloaded from the U.S. Spending website, and preprocessed into the data matrix. During this process, a new category will be added for ”county size”. Each county found in the data set will have its population recorded, as listed on the 2020 U.S. Population Census, which is the most official updated population record. After every county’s exact population is recorded, another new category will be created to record the population range. For this, every recorded population will be divided into equal sized categories, which are created based on the spread of different population numbers present. Once that category has been recorded, the apriori method will be applied to the data matrix, specifically examining the categories of awarding agency, award type, recipient, county name, and county size range. Once this is complete, the most frequent combinations will be examined to find different patterns in different counties and county size ranges.

After examining the rules generated from the apriori method, and determining the strength of the relations, the outliers discovered in the preprocessing step will be examined in order to determiner what characteristics define them over other transactions. For this, anomaly detection will be employed to further prune any values that are significantly higher or lower than the rest of the set. To do this, the IQR method will be used in context of the total obligations directed to the state of Georgia to confirm the flagging of these transactions as anomalies. Once we confirm the anomalies, we will use an isolation forest, where we will randomly select various features of each transaction, with the goal of determining which features cause the anomalies to happen.

IV. Timeline

After the proposal is completed, the planned timeline will take into account three major milestone dates: M2 on March 5, M3 on April 2, and M4 on May 3. For M2, the goal is to completely implement the first data mining technique, which being anomaly detection, and to begin to implement the isolation forest. To accomplish this, preprocessing and data transformation will need to be complete before the data can be mined. The first half of M2 will be focused on accomplishing these tasks, along with setting up the data matrix within Python. At the same time, the preprocessing of the data and scripting of the apriori method will begin. The goal is to complete the initial setup of the data in the first two weeks, then complete and record the results of the apriori method in the second two weeks before March 5. After M2 is completed, the next major goal for M3 will be to implement the isolation forest and use it on the data matrix. The plan is to dedicate the first half of the milestone to scripting and debugging and to use the second half to run the data matrix through it and have a visualization of the resulting sets of data ready for interpretation. The last milestone, M4, will be dedicated to finalizing the interpretation and compiling it in the final report, ready to be presented to the wide audience. There will be 2 anticipated challenges throughout this process. The first is the time it will take to transfer the data into a data matrix for M2. There are over 1400 entries, which will be time consuming to accomplish. Despite this, by doing this in tandem with scripting the apriori method, and by pacing the creation in the span of two weeks, the matrix creation will be ready in time to be used. Secondly, with the sheer volume of transactions and recipients, it may be difficult to generate meaningful sets of common groups of characteristics. There is plenty of room for error here, since it takes more effort and analysis to match strings over integers, and the breadth of data makes this margin of error wider. However, given plenty of time to implement and debug the scripting, the accuracy of the analysis by the program should be close to guarantee.

V. M2

Milestone 2 aims to answer the question: ”What combinations of awarding agencies, recipients, and funding types frequently co-occur in different regions of Georgia?” It examines records of every federal transaction made by the Federal government to Georgia, with a threshold of 20,000,000 dollars, within fiscal year 2025. The goal is to perform apriori to find out what different regions of Georgia (divided by population range) tend to receive the most in federal spending.

VI. Exploratory Data Analysis and Preprocessing

When downloading the data from USASpending.gov, two sets of tables were received, one for contracts and one for grants and direct payments. Since contracts and grants work differently from each other, each table records different types of data. Despite this, both data sets contain the information we need to look for frequent combinations of awarding agencies and recipients in different regions, those being the awarding agency, recipient name, recipient county, and the transaction type. Even though these points of data are under different names, they can be combined under the same name.

We further checked the quality of the data, specifically looking for any missing values or duplicate transaction IDs. There appears to be some missing data, specifically several IDs that are null values. All of these missing ID values are linked to transactions listed as ”Multiple Recipients”. These are transactions, such as Social Security, that are delivered to a collective of individuals whose information is protected

by law, meaning these transactions are not actual null values. When the data is transformed for mining, these values will be replaced with filler ID values to mark each individual transaction.

Next, we checked the number of unique counties present in the data to determine if there was enough variety in county sizes to warrant exploring them as size ratios. There appeared

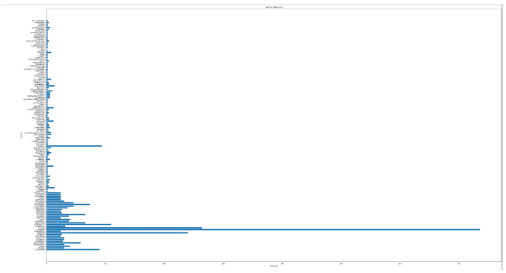

Fig. 3. Screenshot of the generated frequency graph for each county. There displayed 138 counties, however some were located outside of the state.

to be 138 counties, with the majority being smaller counties with a small number of transactions, while a hand full of larger counties received a larger number of transactions. This is perfect for determining what agencies have patterns of funding different sized counties. However, when exploring the counties even further, it was discovered that many of the counties listed were from different states. Many of these recipients have primary locations out of Georgia, but receive Georgia funding for Georgia locations. During preprocessing, we used the zip code listed on USASpending.gov to swap these counties out for their appropriate county. This left only seven transactions

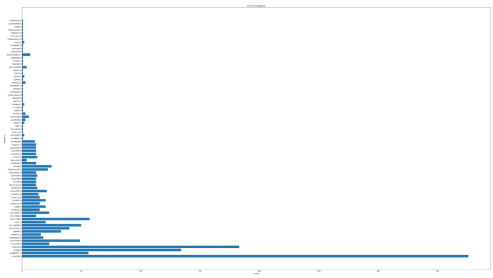

Fig. 4. Screenshot of the generated frequency graph for each county after the counties outside of Georgia were swapped.

without any listed Georgia zip code. With no way of retrieving the Georgia location for these seven recipients, they would only be noise in the algorithm, and were thus removed from the data.

Lastly, before preprocessing occurred, a box and whisker plot was generated to examine the number of outlier transaction values present in the set. The graph revealed that the top 25 percent of transactions were flagged as outlier values. The chances of the top fourth of the data being anomaly transactions was slim, so we needed to determine whether or not there were any anomaly transactions. To do this, a bar graph was generated that showed the total value

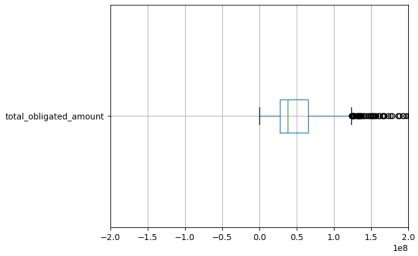

Fig. 5. Screenshot of the generated box and whisker plot for the total obligations.

of each transaction, from ascending order. This revealed a distribution with a gradual increase, and a spike only on the top four transactions, which represented transactions with high-support combinations. From this, it was decided that these transactions should not be removed from the table. There were three transactions marked with total obligations of zero. These are the only three transactions marked as ”loans”. Since loans are requested by the recipient and are paid back to the government, and these transactions would be naturally pruned by the algorithm, the three loans were removed from the data.

VII. Transformation

After exploring the data, almost every necessary piece of data was present. The only column that needed to be added was the county population ranges of each transaction. To do this, another column was added, which recorded the exact size of each county, as listed in the most recent US Census from 2020. From here, we wanted to record the populations into equal sized bins, since we care more about common transaction attributes that relate to counties with similar attributes, rather than the counties themselves. After looking at the distribution of county populations, five bins were created: population 0 - 100,000, population 100,000 - 200,000, population 200,000 - 750,000, 750,000 - 950,000, and populations above 950,000. The spread of these bins are incredibly uneven, however this is due to the wide gap between population sizes that jump from 297595 straight to 764382. This makes the bin a similar size to the rest of the bins.

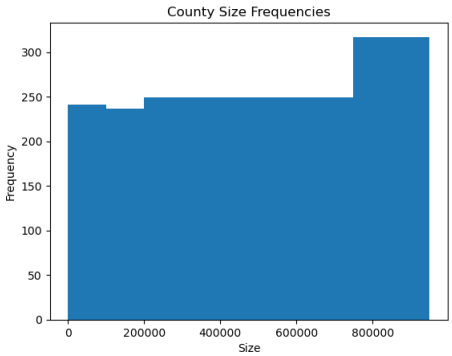

Fig. 6. Screenshot of the generated histogram displaying the frequency for every size ratio bin.

Once this piece of data was added, the final step before apriori is implemented is to list every transaction in a data matrix that can be run in apriori, containing only the relevant columns of data: awarding agency name, recipient name, recipient county name, county size ratio, and award type description. Each row of the matrix represents one transaction labeled with a unique ID from 0 - 1473.

VIII. Data Mining

Once the data was combined into one excel file, and the county size and size ratios were added, they were placed into a 2D matrix, which would be the format used to generate the results of the apriori algorithm. Each row already represented a transaction, so the characteristics of each row were appended onto a list of strings, which would represent one transaction, then tat list was appended into the 2D matrix.

This matrix was then inserted into the apriori algorithm. We began by generating the frequent sets of data. A low support value of 0.01 was used due to the large number of recipients combined with the low number of awarding agencies. With a large number of transactions, even combinations that show up in 20 transactions will display significant information.

Within the collection of frequent sets of data, these are the main patterns that are being discovered:

- Direct payments from the Social Security Administration dominate the most common sets for every county ratio, excluding size 750,000 - 950,000. The administration appears commonly with size 0 - 100,000 with support value of 0.091, size 100,000 - 200,000 with support value of 0.098, size 200,000 - 750,000 with support value of 0.084, and size ¿950,000 with support value of 0.05. While these support values may appear small, they reveal large amounts of social security funding in all of these regions, ranging from 50 - 150 transactions each.
- Regions of size 750,000 - 950,000 appear the most in receiving delivery orders from the Department of Defense, with support of 0.039. This is due to the presence of Lockheed Martin Corp in Cobb County, which is a top recipient of contracts, and therefore skews the award type frequency in this area. The only other non-social security awards that frequently appear in this region are awards made to Emory University in DeKalb, with delivery order contracts appearing at a support value of 0.021 and cooperative agreements appearing at support value of 0.016.
- Only larger regions appear to receive different grant types from a larger variety of agencies. Of the sets with relevant support, Fulton is the only county to receive block grants (Department of Health and Human services with support value 0.023) or formula grants (Department of Education at support value of 0.016, Department of Transportation at support value 0.035, and Department of Agriculture at support value 0.011).
- The Department of Veterans Affairs appears commonly in all regions, however unlike social security, they appear more frequently in smaller regions (Size 100,000 - 200,000 at support value 0.022 and 200,000 - 750,000 at support value 0.029, specifically in regions closer to size 250,000).

The association rules were then generated to help support the common sets found initially. A confidence threshold of 0.8 was used to filter out any rules that would exist due to coincidence. The goal is to discover natural patterns of funding that stay consistent over time.

Hundreds of rules were found that helped support the validity of the frequent sets. Larger counties had a more variety of awarding agencies and funding types compared to more rural areas, with rules displaying confidence that ranged from 0.8 - 1, with lift values between 3 and 4. This is expected, as larger counties with large cities tend to house the majority of larger businesses and government organizations. Smaller counties had almost exclusive frequency with the Social Security Administration and the Department of Veterans Affairs. Each rule that displayed one of these two agencies given one of the smaller counties had a confidence value of 1, and a lift value averaging 6. This reveals the sheer amount of individuals in Georgia that receive government benefits related to retirement programs, and furthermore reveals the number of veterans in smaller areas that benefit from government support, which is a surprising metric compared to urban areas.

IX. Interpretation

When examining all of the major findings from the frequent sets and rules, the majority of findings were trivial, and simply suggest common geographical patterns or common knowledge. The biggest pattern discovered was the large amount of social security transactions distributed in every region of Georgia. This suggests that there are a high amount of retired individuals and families within the state that receive social security or medicaid. The large amount of veterans affairs transactions in smaller counties might suggest a large amount of veterans who receive benefits from multiple government programs. However, without comparing these results to another state, such as Alabama, there is no way of knowing if these numbers are especially high compared to other areas in the country, and without the public ability to see every individual who collects these benefits, there is no way of determining how many of these transactions are legitimate and how many are fraudulent. This discovery does not reveal any differences among Georgia regions in regards to funding patterns.

The large variety of transactions in large regions, such as Fulton County and DeKalb county, is a relevant pattern, however it simply reaffirms that larger businesses and government organizations are found in more urban areas with lots of infrastructure and high population. It is no surprise that they would skew the support values for sets regarding awarding agencies and county sizes. Another factor that appears to skew the support for frequent sets is the existence of military-based operations, such as Lockheed and those found in Warner Robins. Both exist in different sized regions, however they receive a lot of Georgia's federal funding through various contracts. This does not help determine the difference between

different sized regions, rather it reaffirms the existence of important industries, such as military-based businesses, in larger regions.

X. Limitations

While the rules generated by the apriori algorithm are relevant with strong support, the sheer volume of transactions and different recipients makes it difficult to find patterns that are not obvious. The descriptions of each transaction vary heavily, making it nearly impossible to use them as a factor in finding frequent sets of transaction characteristics among different regions. Furthermore, without examining other states in the same manner, it is difficult to display the weight of the significant rules found here. In the future, this same algorithm should be performed in other states of similar size in order to compare their sets. This would better reveal of the high number of social security transactions is exclusive to Georgia, or if they are the largest distribution of federal funds in the country.

Another limitation that may have effected the results is the categories used to create the sets. The county names were kept in addition to their sizes in order to help explain common sets further. However, with the imbalance in counties per size ratio, some sets appear as a consequence of county characteristics being paired with their names. Furthermore, certain recipients only receive specific types of funds from specific agencies. While the less frequent recipients were naturally filtered out, this lead to frequent sets appearing with identical information, but different support values. This was mitigated by favoring certain category combinations over others, such as awarding agencies paired with county sizes, however it is possible that this could have skewed the results and led to the large amount of trivial patterns.

XI. Updated plans for M3

The original plan was to employ anomaly detection using an isolation forest to explain the causes of abnormally large transaction values. However, when exploring outliers in this milestone, it revealed that the largest transactions did not have abnormal characteristics when compared to other transactions. Furthermore, I want to shift into a different approach that may reveal more surprising information than the frequency sets generated by the apriori method. I am shifting my focus onto comparing the characteristics of the recipients themselves as opposed to the geographic regions they appear in. To do this, I will be using K-Means to cluster recipients based on the total amount of funds they received in 2025, the amount of funds dedicated to disaster relief, the amount of awards they received, and the amount of agencies that fund them. Performing this method will help answer the question, ”What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?”

XII. M3

XIII. Implementation Status

As of M3, the main two techniques used for data mining are close to fully implemented. First, the apriori method from M2 is finished and generating data tables. For M3, apriori was used four times:

- 1. Finding common sets of counties, county sizes, agencies, and transaction types.
- 2. Finding reoccurring patterns in combinations of agencies, business types, and counties.
- 3. Finding common combinations of words in transaction descriptions.
- 4. Finding interesting relationships between agencies and recipients.

The first two approaches use a support value of 0.01 and a minimum confidence of 0.7, which is done to find non-trivial patterns in a large dataset of multiple varying factors. Finding common words in transaction descriptions uses a support value of 0.06, which allows non-trivial words, such as ”the” to appear in the table. Finally, finding relations between agencies and recipients uses a very low support value of 0.006, which allows us to see relations that only appear once, which is vital information to analyze. All of these approaches are implemented and working as intended.

Second, k-means clustering is used to find natural groupings of transaction recipients based on obligations and the number of contracts they receive. To do this, each recipient becomes represented by a coordinate based on the two factors and placed in a data frame. Based on the elbow method, we started at 3 clusters, but increased this to 5 to further distinguish members of the largest cluster apart. Currently this method works as intended and is producing meaningful results.

XIV. Analytical Choices

The results of the first run of apriori demonstrated mostly trivial patterns that do not amount to any new information. Due to this, the primary goal for analysis is to discover surprising results over trivial results. To do this, we have chosen to perform a deeper exploration of the apriori method, using different parameters and comparing the results of each run. Each set of parameters will explore a smaller amount of variables in order to try and draw out those surprising results. Specifically, we aim to explore both the relationships between agencies and recipients (both as individual entities and as business types), and the descriptions listed for each transaction. We believe that narrowing the scope of the quantity of analyzed variables will allow us to observe more specific trends that we would not have seen under the conditions of the initial run.

In addition to variations of apriori, we chose to include k-means clustering as another frame of comparison. In addition to discovering more surprising results, we want to directly compare clustering rules to association rules to see which method works best on analyzing this specific set of transactional data. While the nature of the transactions and their sets of classifying variables may seem appropriate for association, examining specific parameters on an individual basis may make clustering a better method for discovering hidden patterns. Grouping recipients by total obligations and total awards received should not only answer out remaining

discovery question, but also allow us to compare the flow of money in addition to the characteristics of the transactions.

XV. Results: Apriori

While the three new sets of parameters did begin to reveal more surprising results, the majority of the results only confirmed what we previously found with the original set. However, as we limited the amount of variables in each run of apriori, we began to see more surprising results.

XVI. Co-occurrence between Awarding Agency, Business Type, and County

Within the frequent sets between awarding agency, business type, and county, there were some surprising patterns, but the data mostly confirmed what we already know. Larger counties, such as Fulton, appear in a bigger variety of rules than smaller counties due to the higher concentration of businesses found within them, while smaller counties appear in less rules and exclusively appear at high support values given that the funding agency is the Social Security Administration. Both the Department of Defense and Corporate companies within Fulton and Cobb exclusively co-occur at support values ranging from 0.03 with a confidence of 0.76. less frequent departments, such as the Department of Transportation and the Department of Agriculture, all exclusively support State Governments found in Fulton County. These results only demonstrate the specialized nature of each federal department in who they fund, which is a trivial metric like the results of the original run.

There are two surprising results found. First, the Department of Veterans Affairs appears with INDIVIDUAL at a high support of 0.07, and a high confidence of 0.95, further reinforcing the high appearance of this department in various low population counties. This can be due to a high concentration of veterans in smaller counties or a low support of veterans in higher counties. This is unlikely to be ill-gotten funding due to the previous discovered variety of veterans assistance in various counties. Therefore, this could be an indication of the rising need for veterans support across the state. Second, the Department of Health and Human services frequently co-occurs in multiple business types, those being NONPROFIT WITH 501C3 IRS STATUS (support of 0.03 and confidence of 0.8), Corporate Non-Tax Exempt (support of 0.02 and confidence of 0.74), and Private Education Institutions (support of 0.01 and confidence of 1). Since there is a Georgia Department of Public Health and a Georgia Department of Human Service, it is surprising to not see frequent co-occurrences between them and the federal department. Furthermore, it is surprising to see frequent co-occurrences with DeKalb, and not Fulton County, however DeKalb is home to many large recipients of federal funding, such as Emory University, which most likely contributes to this discovery. Even so, it is surprising to see these companies receive more funding than the state government.

XVII. Frequently co-occurring words in transaction descriptions

Despite only using one parameter, the descriptions of the transaction purposes did not reveal any surprising information we did not see in the original run, but rather revealed confirmation of what we saw before. There is only one set of meaningful words that frequently occur, those being ”Benefits”, ”Aggregate”, ”(RI)”, ”Retirement”, and ”Insurance”, which all co-occur with each other at support value of 0.37. These are all words that are used to describe transactions from the Social Security Administration, which reinforces their established rank of highest funding agency. This could also reveal the lack of any common interests among the awarding agencies and a nature of specific goals for each transaction, however this finding only shows a lack of patterns that we are looking for.

XVIII. Relations between Funding Agency and Recipient

Unlike the results of the other apriori runs, we saw more surprising results. There appears to be a sizable difference between recipients funded by one agency vs multiple agencies, with there being more recipients with exclusive relations with a specific agency. This is not surprising on its own, however it also appears that the recipients who receive more than one transaction are the ones with exclusive relations. There are a hand full of recipients who have received transactions from multiple agencies, all of which are once of three kinds of businesses: a Georgia Agency, large university, or miscellaneous business who has received one transaction at some point from one of the leading agencies in transactions, those being the Department of Defense, Department of Health and Human Services, and Department of Education. Recipients who have transactions with multiple agencies due to one transaction don't have a high amount of collective transactions and become pruned when raising the minimum support value. The best way to interpret this is that the recipients with lower transaction counts are those who request funding from a specific agency for a specific purpose or project, while those who receive multiple transactions are mostly businesses who might operate off of the funds of one specific agency who funds them. While this may not be surprising, it is interesting to see the way the results shape this rule, and it provides more interesting and surprising rules than the runs that use multiple parameters.

XIX. Results: K-Means Clustering

Similar to the apriori runs, we used k-means to limit the amount of variables to the recipients, and it began to provide surprising and useful results. We performed the elbow method to determine the initial amount of clusters, which revealed 3, however through multiple iterations, we divided the recipients further into five clusters. These clusters can be categorized by these descriptions:

- Group 0: Made up of seven recipients, most of which are state level departments. This category is defined due to

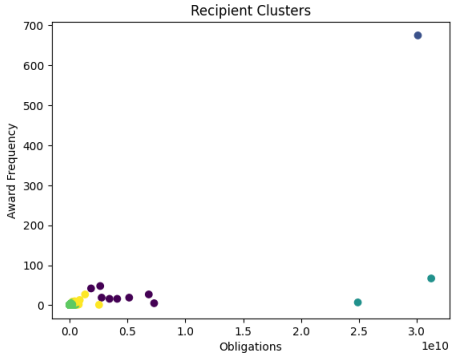

Fig. 7. Graph displaying every recipient and which cluster they belong to.

the large number of transactions made to these departments, which makes sense since the state departments further distribute the funds to other local recipients. This group receives a large number of transactions and dollars compared to the rest of the recipients.

- Group 1: Represented by ”Multiple Recipients”, which makes up those protected individuals funded by social programs from entities like the Social Security Administration. These individuals collectively make up the recipient with the largest number of awards, and one of the largest amounts of money.
- Group 2: Houses Lockheed Martin Corp and the Georgia Department of Community Health, both of which are the other two highest recipients of federal funds. They both received less than 100 contracts over the original threshold of 20,000,000, and therefore do not join group 1 with the multiple recipients of social funds, but still receive enough to create their own category. What is surprising is that Lockheed Martin receives more federal funds than the individuals on social programs. This is an interesting metric that could be interpreted many ways, however without comparing this metric to other fiscal years, it is hard to tell if this is coincidence or not.
- Group 3: The smaller businesses who have received less than 10 contracts, and do not receive as much funding as the rest of the recipients. Most of these recipients have only received 1 transaction, and were pruned from the list of frequent sets done by association. These are the recipients who most likely do not operate mostly on federal funds, however have received grants for specific projects. This category also houses some universities and state departments who have received a minimal amount of contracts compared to their like recipients, and therefore did not make the category with them.
- Group 4: A miscellaneous category. These are recipients who have either received a large amount of funding compared to the low amount of transactions they have received, such as Peraton Inc., or have a low amount of federal funding compared to the high amount of contracts they have received, such as Emory University. There are no real connecting features that define these recipients together, however they appear to be either state departments or universities who didn’t receive enough to be in Group 0, or businesses who received too much to be in Group 3.

XX. Interpretation after M3

After performing apriori under several conditions, as well as comparing it to k-means results, we have clear answers to our discovery questions.

First we asked ”What combinations of awarding agencies, recipients, and award types frequently occur in different sized regions in Georgia?” We found that larger regions generally hold the largest state government programs, military industries, and universities that receive the majority of federal funding. Furthermore, we found that the majority of frequent recipients across the board only co-occur exclusively with one agency, and those who co-occur with multiple agencies do not receive many transactions. Smaller regions are less likely to receive funding from anywhere other than social security. While some of these findings are surprising to see visualized, they are not out of the ordinary.

Second, we asked ”What natural groupings can be found among federal funding recipients based on the characteristics of their transactions?” The largest found group of recipients is the group of small businesses who receive less than 10 transactions, who most likely receive sparing transactions for specific purposes. On the other side of the spectrum is the group of state facilities who receive the majority of funding and receive a frequent amount of transactions. Beyond this is two groups, one held exclusively by individuals who receive social programs, and one held by Lockheed Martin and the Georgia Department of Community Health, who are also in the top three recipients in transactions and funding, however they do not receive as many transactions as the collective of individuals on social programs. Lastly, there is a miscellaneous category for larger firms who did not receive enough to be in the state firm category, or smaller businesses who received too much to be in their respective category.

For the comparison goals listed in the M3 milestone, apriori demonstrated more surprising results when the frequent sets were made up of a smaller amount of variables. This means that it is more effective to examine one parameter based on other variables rather than examining combinations of several variables. With the sheer quantity of transactions, and the sheer variety in transaction details, examining one of those details alone shows more focus and reveals a lot more than the broader scope. This is confirmed once more in the results of k-means, which examined only the characteristics that define the recipient in the given transaction. While the results were not out of the ordinary, it is surprising to see the natural clustering of these recipients and how they compare to each other rather than their individual relations with other qualitative variables.

XXI. Further Limitations

Under the new parameters, the limitations of our preset transaction characteristic thresholds have begun to appear. We

chose to only examine transactions with a minimum obligation of 20,000,000, a decision to reduce dimensionality and make the data more manageable in the given time and resources. This may help in those regards, however there is a chance that this limitation has skewed the data in major ways. For example, there is a significant chance that some of the smaller businesses receive more transactions than is present in the current data set that are not being accounted for. In addition to this, we mentioned that Lockheed Martin received more total obligations than the collective of individuals on social programs. This could be a real possibility given the state of current events in regards to military action, however there is no way of telling if this is a legitimate metric without the total picture of transactions.

We also chose to limit out scope to fiscal year 2025, a decision made for reducing dimensionality, and also to limit the amount of skewed data that may come from any abnormal fiscal years. However, this only increases the amount of trivial data, for without a direct comparison to other fiscal years, it is hard to validate the results of our study. Furthermore, without comparing the results to another state, we cannot tell how surprising our results are. However, since the results of this study are surprising, but not out of the ordinary, they can be used as a direct comparison to other states or other fiscal years to judge the surprising nature of other data sets.

XXII. Updated plans for M4

Every planned technique has been fully implemented, however the code and visuals for the notebook needs to be polished as a whole within the M4 time period. With large tables being the biggest and most relevant visuals, I plan to discover ways to strengthen the quality of these visuals. Furthermore, I also want to add anomaly detection to the study. While the preprocessing portion of the study revealed that there were no real defining characteristics of transactions flagged as an outlier, I want to perform an anomaly detection to confirm this to be the case. I also want to collect data sets from different fiscal years in Georgia, then perform k-means clustering on them in order to directly compare results between each fiscal year, with the goal of seeing how normal or abnormal our current results are. While we have clear answers to our discovery questions, these final tasks should help validate our findings and interpretations.

References

[1] ”State Profile: Georgia.” *USAspending.gov*, https://www.usaspending.gov/state/Georgia. Accessed January 15,2026.